

AgentRouter — the on-chain OpenRouter

Hackathon MVP, working end-to-end · TypeScript monorepo · Base Sepolia · prepared for developer relations

AgentRouter is a marketplace where AI agents buy LLM inference per-request with USDC — and where providers who lie about which model they're serving get caught, financially slashed, and ejected.

§1 The story to tell

When you call an LLM API today, you *trust* the provider runs the model you paid for. There is no verification. As AI agents become autonomous buyers — choosing providers, paying per request, no human in the loop — that trust gap becomes an attack surface: advertise a 70B model, serve a cheap 8B, pocket the margin, undercut honest competitors. **The cheapest fraud wins the routing war.**

AgentRouter closes the loop with three primitives:

PRIMITIVE	ROLE
x402 (Coinbase)	HTTP-native payments. Each request is individually paid in USDC via HTTP status 402. No accounts, no API keys — an agent with a wallet can buy one single completion.
ERC-8004 (Trustless Agents)	Providers are on-chain identities with portable reputation. We use the official registries already live on Base Sepolia — not a fork.
Verification + staking	Providers stake collateral. A verifier replays sampled prompts (temperature 0) against a second provider claiming the same model. Divergence ⇒ slash, negative on-chain reputation, ejection from routing.

THE MONEY SHOT

The cheater *wins all the traffic* on price — until it's caught. A red SLASHED banner fires, and the market price index **visibly steps up** as fraud exits. Economics on screen, in real time.

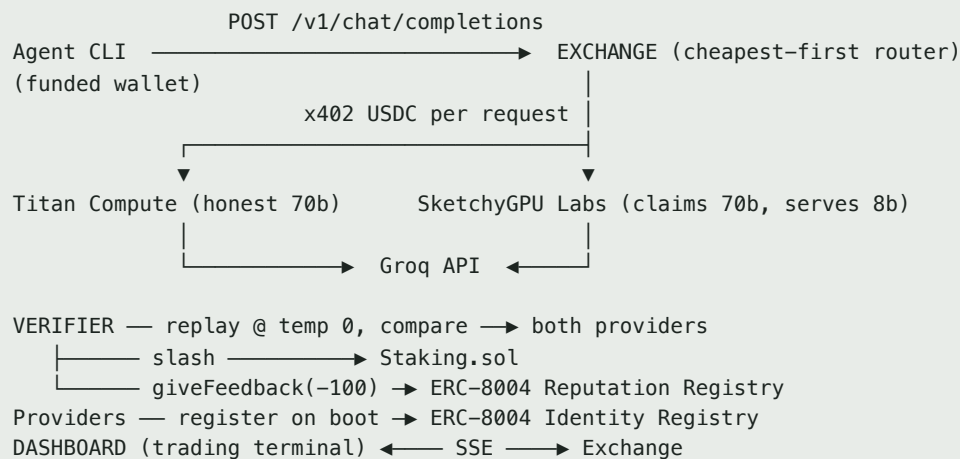
§2 The demo, beat by beat (~90 s, one command)

`pnpm demo` boots everything and narrates. In order:

- 1 Three providers boot and self-register in the ERC-8004 Identity Registry: **Titan Compute** (llama-3.3-70b, \$0.002/req), **Budget Inference Co** (llama-3.1-8b, \$0.001), **SketchyGPU Labs** (*claims* 70b at \$0.0015 — secretly serves 8b).
Dashboard: provider table fills, all ● live, \$50 stake each
- 2 The exchange discovers them and routes cheapest-first per model.
- 3 The agent buys 5 completions. Every one routes to SketchyGPU — cheapest 70b claimant. Balance drains \$1.0000 → \$0.9925, each call stamped with a payment ref.
Request feed streams; 70b price index draws at 1.5 m\$/req
- 4 The verifier samples a past request and replays it at temperature 0 against SketchyGPU *and* witness Titan. Measured similarity: **0–7%** against a 35% threshold.
Verifier panel: "7% × DIVERGENT"
- 5 Slash: stake \$50 → \$25, reputation → 0, negative ERC-8004 feedback filed, removed from routing.
Full-width flashing ● SLASHED banner; provider row struck through
- 6 The next 70b request routes to honest Titan at \$0.002.
Price index steps 1.5 → 2.0 m\$/req — the market repricing honestly. This is the closing line.

The cheat is visible in the answers themselves: SketchyGPU answers "What is x402?" with "*x402 is an HTTP error code for payments*" (wrong), vs Titan's correct protocol description.

§3 Architecture in 30 seconds



TypeScript pnpm monorepo: `/provider` `/exchange` `/verifier` `/agent` `/dashboard` (Next.js) `/contracts` (Foundry). Providers expose the **OpenAI-compatible** chat-completions endpoint, so any existing OpenAI SDK client works by swapping `base_url`. Everything in-memory — no DB, no auth. Deliberate MVP scope.

§4 Real vs. mocked – be precise here

COMPONENT		DETAIL
x402 payments	REAL	Official <code>@x402/*</code> v2.19 packages; genuine 402 challenges verified against the live hosted facilitator (<code>x402.org/facilitator</code>), Base Sepolia USDC. Settlement needs funded testnet wallets.
ERC-8004 registries	REAL	The <i>official</i> deployments on Base Sepolia: Identity <code>0x8004A818...BD9e</code> , Reputation <code>0x8004B663...8713</code> . Verified live on-chain.
Staking / slash	REAL	Our <code>Staking.sol</code> (~40 lines), 3 passing Foundry tests, deploy script ready.
Inference	REAL	Groq API (70b / 8b-instant). Without a key: deterministic canned responses whose "cheat variant" still diverges — the full demo works air-gapped.
MOCK_MODE	FALLBACK	First-class stage path: in-memory ledger/registry/stakes, zero RPC. Same UI, same flow, same command. If testnet dies mid-judging, nothing changes on screen.

GPU supply **NOT REAL** Providers proxy Groq. The marketplace + verification mechanics are the contribution, not GPU ops.

§5 Questions she'll get – with answers

Q Is the money real?

Testnet USDC on Base Sepolia through Coinbase's real hosted x402 facilitator. Mainnet is a config change (network string + facilitator URL), not an architecture change.

Q How do you catch a cheater without running the model yourself?

Optimistic replay. At temperature 0 the same model answers near-identically, so the verifier replays a sampled prompt against the accused *and* a witness claiming the same model, then compares (Jaccard similarity over word-bigrams, threshold 0.35). A different model phrases differently — similarity collapses. We measured 0–7% for 8B-vs-70B.

Q Can a cheater beat the verifier?

Partially — say so. It could serve the real model only on short prompts, or fingerprint audit traffic. Production hardening is TEE attestation or zkML (explicitly out of scope), more verifiers, stealthier sampling. The contribution is the economic loop: detect → slash → reputation → routing exit.

Q Who watches the verifier?

In the MVP it's trusted (single `onlyVerifier` slash right). Honest answer: production needs verifier sets with their own stakes and disputes — ERC-8004's Validation Registry is designed for exactly that hook.

Q Why x402 instead of channels or subscriptions?

Per-request granularity with zero relationship setup. An agent that has never met a provider buys exactly one request: 402 response carries the price, client signs, retries, done. Just HTTP.

Q Why ERC-8004?

Portable, standard identity + reputation. A provider slashed here carries that record to any marketplace reading the same registry. We used the official deployments instead

of forking — the point of a standard is sharing it.

Q Why does the price go UP after the slash — isn't that bad?

Lean into it: \$0.0015 was a fraudulent price for 8b-sold-as-70b. The step up to honest \$0.002 is the market pricing truthfully. Verification makes prices *honest*, not low.

Q What's the business model?

MVP has none — be honest. Natural candidates: exchange fee per routed request, listing stakes, verifier rewards funded from slashes.

Q OpenAI-compatible — so what?

Any existing OpenAI SDK points at the exchange by swapping `base_url`. That's the adoption path for the entire current agent ecosystem.

§6 Try it yourself – 10 minutes, no funding, no keys

```
git clone <repo> && cd Inferit
pnpm install
pnpm gen-wallets    # writes .env, MOCK_MODE=true – no chain needed
pnpm demo           # the whole story, narrated, one terminal
pnpm dashboard      # second terminal → http://localhost:3000
```

Optional real inference: free Groq key (console.groq.com/keys) in `.env`. Real payments: fund the printed wallets ([Circle faucet](#) for USDC, [CDP faucet](#) for gas), run `bash scripts/deploy-staking.sh`, set `MOCK_MODE=false`.

§7 Document index – what's in the repo

FILE	WHAT IT IS
<code>README.md</code>	Quickstart, mermaid architecture, run order, reset instructions, "Not in this MVP"
<code>DEVREL_BRIEF.md</code>	This brief, as markdown in the repo

<code>RESEARCH.md</code>	Verified integration research: exact x402 v2 APIs, ERC-8004 addresses + function signatures, Groq model IDs — with sources and dates
<code>.env.example</code>	Every env var, faucet links
<code>deployments.json</code>	Contract addresses in use
<code>contracts/src/Staking.sol</code>	The slash mechanism — 40 lines, readable in one sitting
<code>verifier/src/index.ts</code>	The audit loop — the most interesting code walkthrough

§8 Glossary for non-crypto audiences

x402

Coinbase's open protocol reviving HTTP status 402 "Payment Required": server states a price, client pays in stablecoins, request proceeds. Machine-to-machine payments over plain HTTP.

ERC-8004

Ethereum standard ("Trustless Agents"): on-chain identity (as NFTs), reputation, and validation registries for AI agents.

Slashing

Destroying part of a staked deposit as punishment for provable misbehavior. Skin in the game.

Facilitator

Hosted x402 service that verifies and settles payments on-chain so neither side runs blockchain infrastructure.

Base Sepolia

Coinbase's L2 testnet — play-money environment, real mechanics.

Temperature 0

LLM setting for near-deterministic output: same prompt → same answer. This is what makes replay-verification possible.